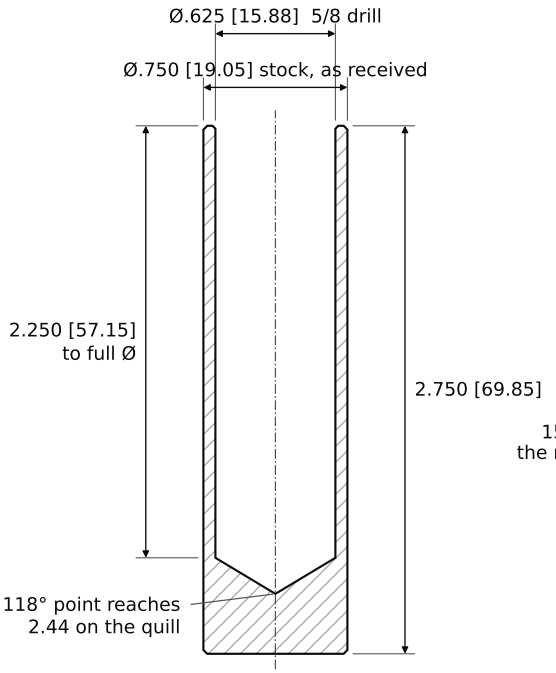
Thin-wall cup and plug for the 22.5 and 30 wt% points of the 4047 spread

The as-made cup (2.750 long, from a 3.000 blank) with a 5/8 hole instead of 1/2, closed by a .250 plug instead of .5625. Every part is cut through the middle; hatching is metal. #152 · #160

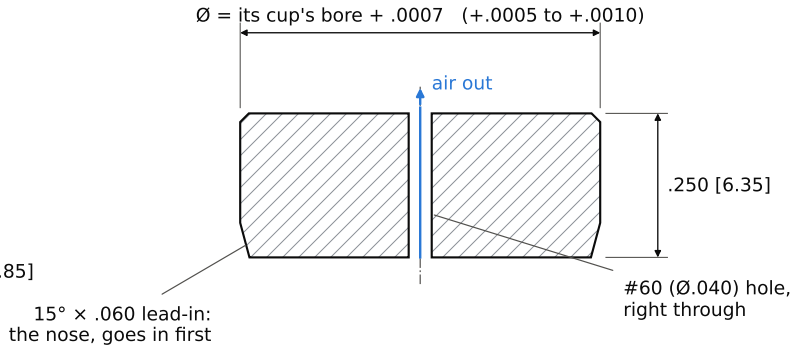
Thin cup 1:1

6063-T52 · one 3.000 blank · ≈22 g
1/16 wall (.0625): check it at the mouth, op 5



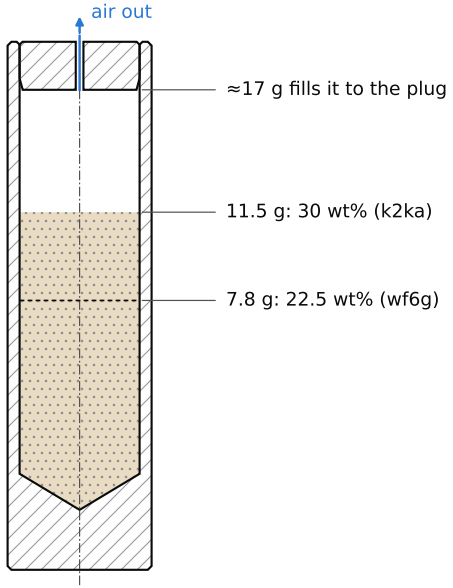
Thin plug 3:1

6063-T52 · one per cup · ≈3.3 g
turned to fit its own cup: not interchangeable



Filled, plug pressed flush 1:1

for reference: stands plug-up in the crucible
levels at 1.60 g/cm³ tapped; weigh, don't fill to a line



HOW TIGHT THE PLUG CAN BE

Over its bore	Cup wall stress, × min. yield	Press force	
+.0005	0.47	0.3–0.4 t	ok
+.0007	**0.65**	**0.4–0.5 t**	**aim**
+.0010	0.93	0.6–0.7 t	max
+.0012	1.12	0.7–0.9 t	too tight

Lamé + von Mises (fit_check() in charge_cad.py), μ 1.05–1.35

THE CUP, FROM ONE 3.000 BLANK (lathe; no parting tool needed)

1 Face both ends of a 3.000 blank to 2.750 ±.005. Break the outside edges .020 × 45°.
2 Center drill. Drill 1/2 to 2.40 on the quill (the pilot).
3 Drill 5/8 to 2.44 on the quill: 2.25 deep to full Ø. Back out every 1/2 to clear chips. Dry or light oil; no water-soluble coolant.
4 Deburr the mouth, .015 max, so the plug starts square.
5 Calipers on the wall at the mouth, 4 places: ≈.060. Note any reading under .050.
6 Measure the bore 1/8 in from the mouth, twice 90° apart (telescoping gauge + mic). Write the average on the record sheet: its plug is made to that number.
7 Flush the bore with 99% IPA, swab, flush again. Dry 80 °C, 30 min. Weigh to 0.01 g.

ITS PLUG, AFTER THE CUP (lathe with a parting tool: CTB shop)

1 Target Ø = its cup's average bore + .0007. Anything +.0005 to +.0010 is fine (table).
2 Face the end of a blank: that face is the nose. Turn the OD to the target over .40. Last passes .002 or less on Ø; let it cool before the last reading.
3 15° lead-in on the nose, .060 long.
4 Spot the center. Drill #60 (.040) .35 deep, so the part-off breaks into it.
5 Part off at .250 ±.005. Deburr the top edge, .015 max. Light shows through the vent.
6 Clean and weigh as for the cup. Bag it with its cup. Don't try it in the cup: it goes in once, after filling. No files, emery, Scotch-Brite, marker or stamps.

FILLING AND CLOSING

1 Weigh the empty cup and its plug together: m.
2 Powder for 22.5 wt% (wf6g) = 0.290 × m. For 30 wt% (k2ka) = 0.429 × m.
3 Tap it down, weigh the filled pair, then press the plug flush, nose first, in the arbor press or a soft-jaw vise (under 0.7 t, table). It stands plug-up from then on.

THIN-WALL CHARGE CUP AND PLUG, Ø3/4 6063

6063-T52, Ø3/4 bar (McMaster 1640T16), the 3.000 sawn blanks · OD as received

Inch [mm] · .XXX ±.005 unless noted · break sharp edges .005–.015

Cups 1:1, plug 3:1 when printed on Letter at 100 %

Rev A · 2026-10-27 · drawn by Claude (AI), PR #172 · checked: _____

Speeds and the record sheet: the process sheets, PR #179 (thin-wall variant)

Source: atomizer-charge/cad/thin_cup_drawing.py